

PROJECT DESIGN PHASE

Solution Architecture

Date	
Team ID	
Project Name	Auto-Ticket Classification using Flow Designer
Maximum Marks	4 Marks

Solution Architecture

The solution architecture describes how the business requirement for simpler school IT ticket submission is connected to the ServiceNow technology components. The design separates the user interface, ticket data, automation logic, notification service, and controlled configuration.

Multi-Tier Solution Architecture Flowchart

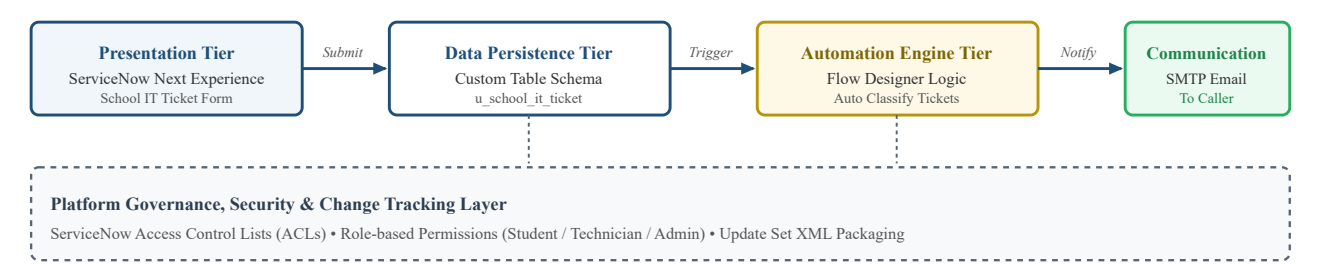


Figure 1: Multi-tier solution architecture connecting UI, schema, Flow Designer, and platform services.

Architecture Flowchart & Diagram Explanation

System Flow & Layer Interaction:

- **Presentation Layer:** Students interact with the ServiceNow Next Experience School IT Ticket form. They input only their identity (Caller) and issue summary (Short Description).
- **Data Persistence Layer:** The custom table u_school_it_ticket captures the record and triggers database events.
- **Automation Layer:** Flow Designer listens asynchronously for record creation, inspects text fields, evaluates conditional keyword rules, and updates dependent fields.
- **Communication Layer:** The platform email engine formats and dispatches the acknowledgment message directly to the requester.
- **Governance & Security Layer:** ServiceNow Access Control Lists (ACLs), role definitions, and Update Sets safeguard schema integrity.

Architecture Components

Component	Role in the solution	Technology
User Interface	Students enter Caller and Short Description and access School IT Tickets.	ServiceNow Next Experience
Ticket Data	Stores Number, Caller, Category, Subcategory, Short Description, Description, State, Assigned Group and Assigned To.	Custom table: u_school_it_ticket
Automation Logic	Checks the Short Description and updates Category and Subcategory for supported issue types.	ServiceNow Flow Designer
Notification	Sends an acknowledgment email to the Caller after ticket submission.	ServiceNow Email / SMTP
Security	Controls access to the custom table and its records.	ServiceNow ACLs and Roles
Change Tracking	Captures custom table definitions, choice fields, flow logic, and email configurations.	ServiceNow Update Sets

Security & Governance Architecture

The solution incorporates security and governance through platform-native controls: Access Control Lists (ACLs) prevent unauthorized record deletion or modification, role-based access restricts configuration capabilities to system administrators, and all changes are strictly encapsulated within an Update Set.

Solution Architecture - Implementation Evidence

Servicenow

All | Favorites | School IT Ticket - TK10001012 | Search | Update | Delete

<

School IT Ticket
TK10001012

▢ TK10001012

Cables System Administrator

Short description
WiFi not working in library

Description

Category Network Assigned Group

Subcategory WiFi Assigned to

State -- None --

Update Delete

Figure 2: School IT Ticket form showing user interface and entry fields.

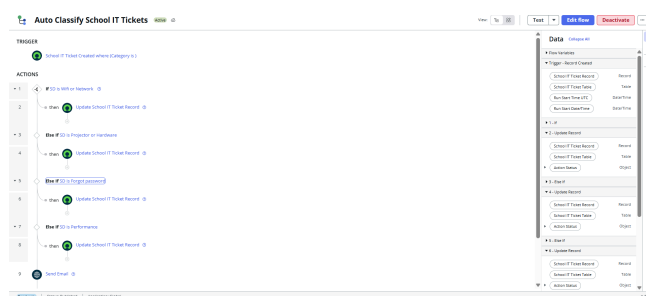


Figure 3: Published Auto Classify School IT Tickets Flow Designer workflow.

services

[Home](#)
[Favorites](#)
[History](#)
[Workspaces](#)

Search IIT Ticket: TKC000000012

Search

[Home](#)
[Home](#)
[Home](#)

TKC000000012

TKC000000012

Category

System Administrator

Best description

TKC000000012

TKC000000012

TKC000000012

TKC000000012

TKC000000012

TKC000000012

TKC000000012

TKC000000012

Category

Network

Assigned Group

TKC000000012

Subcategory

TKC000000012

Assigned To

TKC000000012

Role

None

Update

Delete

Figure 4: School IT Ticket showing Category = Network and Subcategory = Wi-Fi.

[illegible]

Figure 5: Completed Project Update Set with captured customer updates and XML export link.